LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Miscellaneous loading cases, longitudinal cases, fuselage pressurization loads, flap and airbrake loads, spinning and power plants load, fuel systems, freight loads, ground handling guided missile launcher cases
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Miscellaneous loading cases, longitudinal cases, fuselage pressurization loads, flap and airbrake loads, spinning and power plants load, fuel systems, freight loads, ground handling guided missile launcher cases.
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of loading case diagrams; Safety: No physical handling of live components; Instructor supervision: Monitor diagram interpretation accuracy.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary longitudinal loads acting on an aircraft fuselage?
(b) Define fuselage pressurization loads and their effect on the structure.
(c) Identify the structural elements responsible for resisting flap and airbrake loads.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Miscellaneous loading cases definition • Longitudinal load distribution analysis • Fuselage pressurization load mechanisms • Flap and airbrake load characteristics • Spinning and power plant load factors	L & Ds	PPT, IB	15 min
(b)	• Fuel system load integration • Freight load distribution principles • Ground handling and missile launcher cases • Structural response to combined loads • Load factor calculation basics	L & Ds	PPT, PDF	20 min
(c)	• Comparison of static vs dynamic loads • Impact of missile launcher cases on structure • Review of fuel system stress points • Summary of freight load safety limits • Integration of all loading cases	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how fuselage pressurization loads differ from longitudinal cases.
(b) Describe the specific structural challenges posed by guided missile launcher cases.
(c) How do flap and airbrake loads affect the wing structure during flight?

12. Home Assignment and Follow Up	Prepare a brief note on the joining techniques used in aircraft fuselage structures to be discussed in LP 09.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Classification of aircraft parts, types of aircraft fuselage structure, joining techniques, riveted joints, bonded joints, brazing.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min